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A Research Proposal on

"Design & Development of a Comprehensive Digital Nutrition Support & Sustainable Food Intelligence Platform (App) for Nutrition Professionals, Students & Nutrition-Conscious Individuals in Bangladesh"

Scholar's name : Mohima Sarwatun

University name : Chattogram Veterinary & Animal Sciences University

Degree : Bachelor of Science (Hons)

Faculty : Food Science & Technology

Year : 2025

Country : Bangladesh

Submission date : 20/09/2026

INTRODUCTION

The term diet refers to foods and beverages consumed over time in all settings, such as worksites, schools, restaurants, and the home. Diet also often means a specific nutritional plan or eating pattern. Nutrition is the process of consuming, absorbing, and using nutrients from food that are necessary for growth, development & maintenance of life [1].
However, Bangladesh continues to face a complex nutritional burden, where undernutrition, inadequate dietary diversity, micronutrient deficiencies & unhealthy dietary practices coexist. The nutritional challenges also extend to adolescents, young adults & the general adult population. A recent study among university students in Bangladesh reported that only 32.5% of participants had satisfactory nutrition knowledge, while approximately 40% demonstrated high dietary diversity [2]. Among Bangladeshi adults, nutrition literacy has also been found to be significantly associated with healthy eating behavior's, suggesting the importance of accessible & understandable nutrition information in supporting appropriate dietary practices[3]. In addition, nationally representative evidence has reported considerable consumption of unhealthy foods among Bangladeshi adults aged 20–59 years, including frequent consumption of sweet snacks and sugar-sweetened beverages[4]. These findings highlight the continuing need for accessible, practical & evidence-based nutrition support across different population groups.
Although professional nutrition services in Bangladesh are becoming increasingly recognized as an important component of health & disease management but obtaining individualized nutrition support may still be challenging for many individuals. Nutrition professionals need to collect anthropometric measurements & dietary information to assess dietary patterns & nutrition related risk factors, these tasks are performed manually. So the process can become repetitive & time consuming. Nowadays with the increasing use of digital health technologies, various nutrition applications & digital tools have been developed with separate functions[5],[6]. However, the functions required throughout a professional nutrition care workflow are often distributed across different tools rather than being organized within a single specific platform.

Therefore, this proposal aims to develop a comprehensive digital nutrition support & sustainable food intelligence platform(App) in Bangladesh, with separate access & functions for nutrition professionals, students & patients. The proposed platform would allow nutrition professionals to manage a patient's nutritional assessment from the initial appointment, recording anthropometric & dietary information, prepare personalized diet charts digitally, access updated food database & nutrition information, identify appropriate food groups & portions, explore food alternatives.
The completed diet chart could be printed, used as a reference for manual documentation or shared directly with the patient through the application. Patients would be able to access their assigned diet charts, review food & portion information, explore recommended alternatives, maintain follow up records & communicate with the nutrition professional through the platform. Professionals & students would have broader access to structured nutrition, food related information & regularly updated educational resources. The platform would also incorporate sustainable food alternatives by presenting relevant nutritional facts. Artificial intelligence would function as a supporting tool for tasks such as summarizing information, organizing records, & assisting with explanations rather than replacing professional nutritional judgment. By bringing these functions together within one platform, the proposed application is intended to provide a Bangladesh specific digital framework that reduce repetitive administrative & information search tasks, facilitate more organized & accessible nutrition services & nutrition awareness among students & patients.

Research deficiency

i) Existing nutrition apps mainly focus on dietary tracking, assessment or education [5], [6].
ii) Digital dietary assessment tools mainly focus on food intake or diet chart rather than complete professional nutrition care.
iii) Bangladesh has a food composition database, but it is not integrated into a comprehensive nutrition support platform [7].
iv) Personalized nutrition & sustainable food choices are mostly studied separately [8], [9].
v) A Bangladesh specific platform integration of nutrition, affordability & sustainability up remains limited.

Aim of the research

To design a Bangladesh specific Comprehensive Digital Nutrition Support Platform that would integrate digital diet chart, food information hub, sustainable alternative food suggestions, assessment & personalized recommendation, patient follow up, learning & professional update.

Objectives

i) To develop a comprehensive digital system platform for nutrition assessment & dietary pattern analysis.
ii) To provide Bangladesh specific food, nutrition, portion, food group, cost effective alternative-food information.
iii) To facilitate personalized digital diet chart preparation, explanation, modification & delivery.
iv) To enable patient food diary, symptom tracking, follow up & professional communication.
v) To support sustainable food choices, nutrition education awareness among professionals, students & patients, AI-assisted functions & platform usability.

LITERATURE REVIEW

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i) Mobile applications have been used for dietary intake assessment, nutrition education, behaviour improvement, and nutrition monitoring [10].
ii) Dietary mobile applications have shown potential to improve nutritional outcomes, including weight, waist circumference, and energy intake, among adults with chronic diseases [11].
iii) Dietary-record applications can estimate energy and nutrient intake, but validation is important because differences from traditional dietary assessment methods have been reported [12].
iv) Reviews of nutrition applications indicate that individualized dietary recommendations and access to information for healthcare professionals are not consistently available across existing applications [13].
v) Recent evidence suggests that mobile applications may also support healthier and more sustainable dietary choices, although further evidence is needed for different populations and dietary outcomes [14].

Research Questions

1. What nutrition related needs of professionals can be addressed through one digital platform?
2. How can the platform improve information access and reduce time spent on repetitive tasks?
3. How can nutrition assessment, dietary evaluation, food information, and diet-chart preparation be integrated into one platform?
4. How can the platform support personalized diet planning & patient follow up?
5. How relevant & useful is the proposed platform for nutrition professionals?

METHODOLOGICAL FRAMEWORK

1. Approach: The proposed plan will be conducted using mix-method (Quantitative & Qualitative). Quantitative data will be used to evaluate time efficiency, accuracy, usability, and user experience, while qualitative feedback will identify user needs, difficulties, and areas for improvement.

2. Literature Review: At first relevant studies will be reviewed to identify existing nutrition tools/applications, assessment methods, food databases digital diet planning approaches.

3. Requirement Identification: The requirements of nutrition professionals will be identified, focusing on nutrition assessment, food information, diet-chart preparation & patient follow up.

4. Food Database Development: Bangladesh specific food information will be collected & organized according to food groups, nutrients, portions, cost, availability & alternatives.

5. System & App Development: The Application platform will be developed through the following outline:
User Profile → Nutrition Assessment → Dietary Evaluation → Food Information → Personalized Diet Chart → "Why This Diet?" → Food Diary & Symptom Tracking → Patient Follow-up → Sustainability & Educational Modules → Professional Dashboard.

6.Prototype Development: An initial prototype will be developed by integrating the major modules and establishing the connection between assessment, dietary evaluation, food information, and diet chart preparation.

7.Participants and Study Area: The prototype will be tested by selected nutrition professionals to assess: Time efficiency, Usability, Diet-chart accuracy & relevance Follow up usefulness Information organization, User experience.

8.Data Analysis: Quantitative findings will be analyzed using descriptive statistics and appropriate statistical tests. Qualitative feedback will be analyzed thematically and used to improve the prototype.

9.Final Prototype: The evaluated prototype will be refined based on user feedback and performance findings to produce the proposed integrated digital nutrition support platform.

10.Research Time Period: The study may take18 months (1.5 years).
i) Months 1–3: Literature review & user requirement identification.
ii) Months 4–6: Food database development &system framework design.

iii) Months 7–11: Development of the major platform modules.
iv) Months 12–14: System integration &functional prototype development.
v) Months 15–17: Prototype testing, user evaluation and feedback-based modification.
vi) Month 18: Final analysis, documentation and completion of the prototype.

Expected Results

i) A functional integrated nutrition-support prototype.
ii) A Bangladesh-specific food and nutrition database.
iii) Personalized digital diet-chart preparation.
iv) Community-level adoption and market uptake
v) Improved professional workflow and patient follow up.

Potential Challenges

i) Limited availability of reliable Bangladesh-specific food data.
ii) Difficulty in maintaining accurate and updated food information
iii) Challenges in developing accurate nutrition assessment and diet chart functions.
iv) Ensuring data privacy and security.
v) Limited user participation during prototype testing.
vi) Technical challenges in building integrating multiple platform modules.

CONCLUSION

Developing an integrated digital nutrition platform in Bangladesh may be challenging due to data, technical and practical limitations. However, this platform will bring various functions into one system. It will benefit nutrition professionals, students & patients by reducing extra time, improving information access & supporting more effective & faster nutrition services. It will also help to improve nutrition related knowledge and reduce practical difficulties in nutrition care. Ultimately, this research aims to develop an effective platform for more organized and accessible nutrition support.

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